



Oluwatobiloba Oyelowo

Application Security Researcher & Penetration Tester

■ Ibadan, Nigeria · ✉ tobjosh@wearehackerone.com
■ github.com/Oyeetobjosh · ■ bugcrowd.com/Tobjosh

PROFESSIONAL SUMMARY

Results-driven Application Security Researcher specialising in **IDOR/BOLA vulnerability research**, REST/GraphQL API security testing, and OAuth 2.0/JWT exploitation. Proven track record of responsibly disclosing valid security vulnerabilities to top-tier programs including a major global cryptocurrency exchange via Bugcrowd. Experienced in systematic multi-target assessments across fintech, food delivery, developer platforms, and SaaS ecosystems. Builds custom offensive security tooling and maintains an active presence across leading bug bounty platforms.

CORE COMPETENCIES

Vulnerability Classes	Tools & Platforms	Protocols & Tech
IDOR / BOLA · Broken Access Control OAuth 2.0 Flow Attacks · JWT Exploitation API Security Testing · Mass Assignment HTTP Parameter Pollution Information Disclosure · XSS	Burp Suite · Kali Linux Custom Python Tooling · cURL New Relic (APM Monitoring) Browser DevTools · Wireshark Git / GitHub · Postman	REST API · GraphQL · OAuth 2.0 JWT / JWS · WebSocket HTTP/2 · JSON / XML Cookie Auth · Bearer Token Linux / Bash Scripting

SECURITY RESEARCH EXPERIENCE

Independent Security Researcher — Bug Bounty

Bugcrowd & HackerOne Platforms · 2024 – Present · Remote

- Identified and responsibly disclosed valid security vulnerabilities to a **major global cryptocurrency exchange** via Bugcrowd, demonstrating ability to assess high-security fintech targets.
- Conducted vulnerability assessments against programmes operated by **HostGator** (web hosting infrastructure) and other technology companies, identifying access control and configuration weaknesses.
- Performed systematic IDOR/BOLA research across multiple public programmes, specialising in cross-tenant data access, API authentication bypass, and OAuth 2.0 implementation flaws.
- Developed **EYE-door** — a custom 940+ line Python IDOR/BOLA scanner implementing dual-session comparison, Levenshtein/Jaccard similarity diffing, GraphQL node substitution, HTTP Parameter Pollution, and automated method escalation (GET→PUT→DELETE).
- Built **tobjosh-recon** (GitHub) — a powerful recon scanner producing structured output for AI-assisted vulnerability analysis across large attack surfaces.
- Performed OAuth 2.0 flow security testing including authorisation code replay, redirect_uri bypass (8+ variants), scope escalation, and cross-app token confusion attacks.
- Conducted JWT security analysis: payload decoding, algorithm confusion, claim manipulation, and token lifetime exploitation.
- Used **New Relic** APM data and JS bundle mining to identify internal service URLs, undocumented API endpoints, and configuration secrets exposed via client-side assets.
- Assessed multi-actor food delivery platform security across consumer, restaurant, and payment microservices using dual-account BOLA methodology with Burp Suite and Kali Linux.
- Operated under strict responsible disclosure methodology in compliance with Bugcrowd and HackerOne program policies and coordinated vulnerability disclosure (CVD) standards.

Security Tool Developer

Personal Projects — GitHub: github.com/Oyeetobjosh · 2024 – Present

- **EYE-door IDOR Scanner** — Modular detection framework with 6 attack engines, IDHarvest module, Response Graph Mapper, Race Engine, and automated H1/Bugcrowd report generator.

- **tobjosh-recon** — Advanced reconnaissance scanner generating structured .txt intelligence reports optimised for AI-assisted vulnerability analysis.
- **Hacker-Portfolio** — Interactive terminal-style security portfolio built in HTML/JS, demonstrating frontend development capabilities alongside security research.
- Maintained active open-source presence across security tooling repositories with documented methodologies for IDOR/BOLA research.

NOTABLE PROGRAMME ASSESSMENTS

Cryptocurrency Exchange (Bugcrowd)
Identified and responsibly disclosed valid security vulnerabilities via Bugcrowd to one of the world's leading cryptocurrency exchanges. Demonstrated capability to assess high-security, high-value financial platforms.
Web Hosting Infrastructure (Bugcrowd)
Security assessment of web hosting platform infrastructure. Identified access control and configuration vulnerabilities affecting hosting service components.
Developer Platform (Public BB)
52-phase IDOR/BOLA assessment across REST and GraphQL surfaces. Mapped 100+ endpoint/role combinations. Identified internal service URL exposure via client-side JS bundle analysis.
Global Payment Network (Public BB)
Assessed cross-merchant BOLA across OAuth partner flow. Tested authorisation code replay, redirect_uri bypass, and cross-app token confusion on sandbox environment.
Food Delivery Platform (Public BB)
Dual-account API assessment across consumer, restaurant, and payment microservices. JWT payload analysis and cross-user endpoint mapping across 20+ attack surfaces.

EDUCATION & PROFESSIONAL DEVELOPMENT

Bachelor's Degree (In Progress)

University, Nigeria · Expected: 2026

- PortSwigger Web Security Academy — Access Control, API Security, OAuth 2.0, JWT Attacks, Business Logic
- HackerOne Hacker101 — Web Application Security, API Security, Authentication Vulnerabilities
- Self-directed analysis of 250+ disclosed IDOR/BOLA reports on Bugcrowd and HackerOne — pattern extraction and exploit reproduction
- OWASP API Security Top 10 — Applied across real-world bug bounty targets
- Kali Linux hands-on practice — Network scanning, web application testing, tooling

BUG BOUNTY PLATFORMS & PROFILES

Bugcrowd	bugcrowd.com/Tobjosh — Active researcher with valid findings on top-tier programmes
GitHub	github.com/Oyeetobjosh — Security tooling, recon scanner, hacker portfolio

All security research conducted ethically within authorised bug bounty programme boundaries. Programme-specific details withheld in compliance with responsible disclosure and non-disclosure obligations. Detailed engagement reports and references available upon request.